

LinFill-120 Vertical Syringe Filling Station

LinFill Series · Elara Automation GmbH

MQTT Interface Specification

This specification defines the complete MQTT 3.1.1 communication interface of the LinFill-120 Vertical Syringe Filling Station. It is intended for control engineers, SCADA/MES integrators, and Unified Namespace architects who need to connect the station to a supervisory system or data infrastructure.

All information in this document refers to firmware version 1.4.2 and later. Earlier firmware versions may not support all features described here.

Document No.	EA-MI-LF120-EN	Revision	1.0
Release Date	2024-04-12	Language	English (EN)
Manufacturer	Elara Automation GmbH — www.elara-automation.de	Status	Released

© 2024 Elara Automation GmbH. All rights reserved. This document and the information contained herein are the property of Elara Automation GmbH. Reproduction, disclosure, or use without prior written authorisation of Elara Automation GmbH is strictly prohibited. The information is provided in good faith and is believed to be accurate at the time of publication. Elara Automation GmbH reserves the right to modify specifications without notice. Product names and logos are trademarks of their respective owners.

About This Document

This document has been prepared by Elara Automation GmbH and is intended for qualified technical personnel responsible for the installation, commissioning, operation, and maintenance of the equipment described herein. Readers are expected to be familiar with general principles of industrial automation, electrical safety, and network-connected laboratory equipment.

The document is structured as a reference manual. Sections are broadly ordered to follow the typical lifecycle of the equipment — from first delivery through to end-of-life disposal — but individual sections may be read in isolation. Cross-references are provided where information in one section depends on material covered elsewhere.

Elara Automation GmbH maintains a document register at <https://elara-automation.de/documents>. The latest revision of this document may be retrieved from that register using the document number printed on the cover page. Printed copies are uncontrolled; always verify the revision status before use.

Conventions Used in This Document

The following conventions are used throughout:

Notation	Meaning
{serialNumber}	The unit's serial number as printed on the rear label (e.g. EA-PS-2024-00391)
{base}	The site-specific topic or namespace prefix, configured during commissioning
NOTE	Supplementary information that aids understanding but is not safety-critical
CAUTION	Risk of equipment damage or process disruption if instruction is not followed
WARNING	Risk of personal injury if instruction is not followed

Related Documents

Document No.	Title
EA-LC-LF120-EN	LinFill-120 Lifecycle Information Document
EA-LN-SPL01-EN	SPL-01 Line Description & Bill of Materials

1 Introduction

MQTT (Message Queuing Telemetry Transport, ISO/IEC 20922:2016) is a lightweight, publish-subscribe messaging protocol well suited to embedded devices with constrained resources and unreliable network connections. The LinFill-120 acts as an MQTT client and connects to a broker provided by the facility. It does not include an embedded broker; a separate broker (e.g. Eclipse Mosquitto, HiveMQ, EMQX) must be available on the facility network.

The station publishes operational data to three topics and subscribes to one command topic. The topic prefix — referred to as {base} throughout this document — is site-specific and configured during commissioning. There is no default value; the prefix must be agreed between the equipment owner and the system integrator before commissioning and recorded in the site namespace register.

2 Transport and Connection Parameters

The MQTT connection is carried over TCP using the device's 2.4 GHz WiFi interface (IEEE 802.11 b/g/n). The broker port number depends on whether TLS is in use. For development or isolated lab networks, plain TCP on port 1883 is acceptable; for production environments Elara Automation strongly recommends MQTTS on port 8883 with mutual TLS certificate authentication.

Protocol version	MQTT 3.1.1 (ISO/IEC 20922:2016)
Transport	TCP/IP over IEEE 802.11 b/g/n WiFi, 2.4 GHz band
Default port (plain)	1883
Recommended port (TLS)	8883 with TLS 1.2 or higher
Client ID	linfill120-{serialNumber} — e.g. linfill120-EA-LF-2024-00472
Clean Session	false (persistent session; QoS 1 subscriptions survive reconnects)
Keep-alive interval	60 seconds (broker will detect offline device after 90 s)
Reconnect strategy	Exponential back-off starting at 1 s, doubling up to 60 s maximum
Authentication	Username / password pair, set during provisioning. TLS client certificate supported in addition to or instead of password.

2.1 Last Will and Testament

The client registers a Last Will and Testament (LWT) message at connection time. If the station disconnects unexpectedly — due to power loss, WiFi drop, or firmware watchdog reset — the broker will publish the LWT payload to the state topic after the keep-alive timeout expires. Supervisory systems should subscribe to the state topic with a retained-message handler so that the last known state is immediately available on (re)subscribe.

LWT topic	{base}/filling/state
LWT payload	"ERROR"
LWT QoS	1 (at least once)
LWT retain	true

3 Topic Hierarchy

All topics used by the LinFill-120 are relative to the {base} prefix. A typical prefix for a pilot filling line might be *site/lab/line1/linfill120*, but the actual value is entirely site-defined.

Full Topic (relative to {base})	Direction	QoS	Retained?
/filling/state	Device publishes	1	Yes
/filling/cycletime	Device publishes	1	No
/filling/syringeweight	Device publishes	1	No
/filling/cmd	Device subscribes	1	No

4 Published Topics

4.1 {base}/filling/state

The state topic carries a short UTF-8 string describing the current operational mode of the station. It is published on every state change (event-driven) and retained by the broker, meaning a new subscriber immediately receives the most recent value without waiting for the next state change.

Payload value	Meaning	Entry conditions
"IDLE"	Ready; all actuators at rest	After successful HOME; after cycle completion; after STOP; after RESET clears error
"RUNNING"	Fill cycle in progress	START command received while State == IDLE
"ERROR"	Fault latched; motion inhibited	Limit-switch timeout; motor stall; L298N thermal shutdown; LWT published by broker
"HOMING"	Lift performing homing sequence	HOME command received; also entered automatically at power-on

4.2 {base}/filling/cycletime

Published once per cycle, immediately after the state transitions from RUNNING to IDLE following a successful fill cycle. The payload is the cycle duration in seconds, serialised as a decimal string with two decimal places (e.g. 4.73). This topic is not retained.

NOTE: *Cycletime is not published if the cycle is interrupted by a STOP command or if the station faults during a cycle. Supervisory systems should treat absence of a cycletime message as an indicator of an incomplete cycle.*

4.3 {base}/filling/syringeweight

When an external balance is connected to the ESP32 UART, the station publishes a post-fill syringe weight reading to this topic on each cycle completion. The payload is a floating-point value in grams, serialised as a decimal string (e.g. 1.023). From firmware v1.4.2, the published value is the average of three consecutive balance readings taken after the lift has retracted; earlier versions published a single reading.

5 Subscribed Topic

5.1 {base}/filling/cmd

The station subscribes to this topic for incoming commands. The payload is a single UTF-8 string. Commands are case-sensitive; lowercase or mixed-case strings will be silently ignored. Commands received in an incompatible state (e.g. START while RUNNING) are also silently discarded. No acknowledgement or error message is published in response to an invalid or ignored command; supervisory systems should implement timeout monitoring

on the state topic.

Command	Valid when State is	Resulting state sequence	Notes
"START"	IDLE	IDLE → RUNNING → IDLE	Begins fill cycle; cycletime and weight published on completion
"STOP"	RUNNING	RUNNING → IDLE	Halts cycle; lift retracts. Cycletime NOT published.
"HOME"	Any	* → HOMING → IDLE	Should be issued once on power-on before first START
"RESET"	ERROR	ERROR → IDLE	Clears fault flag; does not re-home

6 Security Recommendations

The default configuration (unencrypted TCP, username/password authentication) is adequate for isolated development networks. Before connecting the station to any production or GxP-regulated network, the following additional measures should be considered.

- **TLS encryption:** Enable MQTTS on port 8883. Provision the broker CA certificate onto the ESP32 via the provisioning UI. Consider mutual TLS (client certificate on the device) to authenticate the device to the broker.
- **Access control lists:** Restrict the client ID `linfill120-{serialNumber}` on the broker to publish/subscribe only to `{base}/filling/#` and no other topics.
- **Network segmentation:** Place filling stations on a dedicated IoT VLAN with firewall rules restricting MQTT traffic to the authorised broker IP address only.
- **Credential rotation:** Change device passwords and rotate certificates at least annually, or following any personnel change that had access to the credentials.
- **Broker hardening:** Disable anonymous access on the broker. Log all connect and disconnect events. Set broker-side keep-alive expiry slightly longer than the device keep-alive (90 s is appropriate for a 60 s device keep-alive).

7 Sparkplug B Integration

The LinFill-120 firmware does not implement the Sparkplug B specification natively. If integration into a Sparkplug B Unified Namespace is required, an edge node must bridge the native topics to Sparkplug NBIRTH/DBIRTH/DDATA messages. The recommended metric mapping for such a bridge is shown below.

Native Topic (relative)	Sparkplug Metric Name	Sparkplug Datatype	Notes
/filling/state	State	String	Map to Sparkplug DDATA on change
/filling/cycletime	CycleTime_s	Double	Publish on cycle completion event
/filling/syringeweight	Weight_g	Double	Publish on cycle completion event
/filling/cmd	Cmd	String	Write via Sparkplug DCMD

8 Integration Checklist

The following checklist may be used as a quick commissioning aide-memoire. It does not replace the full OQ protocol.

#	Item	Done?
1	Broker reachable from device subnet; firewall permits TCP 1883 or 8883	■
2	Client ID linfill120- <code>{serialNumber}</code> is unique on the broker	■
3	<code>{base}</code> prefix agreed with namespace owner and recorded	■
4	LWT configured; test by powering off device and confirming ERROR appears on state topic	■
5	SCADA/MES subscribes to <code>/filling/state</code> with retained-message support	■
6	SCADA/MES subscribes to <code>/filling/cycletime</code> and <code>/filling/syringeweight</code>	■
7	All four commands tested; state transitions verified per §5.1	■
8	TLS and ACL configured for production use	■
9	Keep-alive alarm dead-band set to ≥ 90 s in SCADA	■
10	Firmware version confirmed as 1.4.2 or later before go-live	■

9 Technical Support

For integration support, firmware files, or to report unexpected behaviour, contact Elara Automation GmbH using the details below. When raising a support ticket please include the unit serial number, the firmware version visible in the provisioning web interface, and a description of the MQTT broker software and version in use.

Elara Automation GmbH

Brückenstraße 14, 70173 Stuttgart, Germany

Tel: +49 711 4920 0 · Fax: +49 711 4920 99

support@elara-automation.de

www.elara-automation.de

Business hours: Monday–Friday 08:00–17:00 CET